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**“ECOFLEET AI: INTELLIGENT ELECTRIC VEHICLE FLEET
MONITORING AND ROUTE OPTIMIZATION”**

CASE STUDY REPORT

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Full Case Study Analysis Report

Abstract

The global shift toward electric vehicles (EVs) is reshaping logistics, public transport, and delivery operations, but it has also exposed a gap in how fleets are actually managed. Traditional fleet-management software was designed around fuel vehicles and simply repurposed for EVs — it tracks location well but has no intelligence about battery degradation, charging strategy, or energy-aware routing. This report presents EcoFleet AI, an AI-powered platform that continuously monitors battery health via IoT sensors, predicts maintenance needs using machine learning, optimizes charging schedules against real-time electricity pricing and station availability, and recommends energy-efficient routes using live traffic and vehicle performance data. The report analyzes the problem context, identifies stakeholders and their challenges, evaluates the shortcomings of current fleet-management approaches, and proposes a six-module microservices architecture built on sound software engineering principles — modularity, scalability, maintainability, reliability, usability, and security. An Agile/Scrum development methodology with DevOps/MLOps integration is justified as the most suitable approach given the platform's evolving, data-dependent components. Supporting mathematical models are provided for battery state-of-health estimation, charging-cost optimization, and energy-aware route calculation. The analysis concludes that while EcoFleet AI can meaningfully reduce operating costs, extend battery life, and lower emissions, its success depends on data quality, integration with legacy telematics, and sustained investment in model retraining (MLOps) as the fleet and charging ecosystem evolve.

1. Introduction

1.1 Background and Motivation

Transportation accounts for a substantial share of global carbon emissions, and electrification has become one of the most direct levers governments and businesses have to reduce that footprint. Logistics companies, public transit authorities, and delivery services are converting their fleets to EVs not just for environmental compliance but because total cost of ownership over a vehicle's life can be lower than diesel or petrol equivalents — provided the fleet is operated efficiently. That "provided" is doing a lot of work. An EV's economics depend heavily on how well its battery is managed, how intelligently it is charged, and how efficiently its routes are planned — three areas where conventional fleet software offers little more than a map and a fuel-gauge-style battery percentage.

1.2 Why This Problem Needs a Software Engineering Solution

The challenge is not a lack of data — modern EVs and charging stations already generate large volumes of telemetry (battery voltage, temperature, GPS position, charging session logs). The challenge is the absence of a system that turns that raw telemetry into predictive, actionable intelligence, and does so reliably at fleet scale. This is fundamentally a software engineering problem: it requires distributed data ingestion, machine learning pipelines, optimization algorithms, and a maintainable architecture that can scale from a pilot fleet of a few vehicles to thousands, all while remaining secure and usable by non-technical staff.

1.3 Report Structure

This report proceeds through: understanding the problem statement and its scope; identifying stakeholders and the challenges they face; analyzing the limitations of current fleet-management approaches; proposing a software architecture with detailed functional and non-functional requirements; justifying a development methodology; presenting supporting mathematical models for the platform's core algorithms; and discussing expected outcomes, risks, and future enhancements.

2.Understanding the Problem Statement

2.1 Industry Context

Fleet operators are adopting EVs to cut emissions and fuel costs, but managing an EV fleet is operationally different from managing a combustion-engine fleet. Battery health degrades non-linearly with usage patterns, charging must be scheduled around limited station capacity and variable electricity pricing, and route planning must account for range limitations that don't exist with liquid fuel .A petrol vehicle can always find a fuel station; an EV route must be planned so the vehicle never runs out of charge between chargers.

2.2 Problem in Own Words

EV fleet operators today juggle three interconnected concerns — battery health, charging logistics, and route planning — using tools that were never designed to connect them. A conventional GPS tracker shows where a vehicle is, not how much life is left in its battery or whether its planned route will require an unplanned charging detour. This disconnection means small inefficiencies (a battery approaching end-of-life, a poorly timed charging stop, a route that ignores elevation-driven energy cost) compound into missed deliveries, higher operating costs, and unnecessary battery replacements.

2.3 Objectives

1. Build an AI-powered platform for continuous EV fleet performance monitoring.
2. Monitor battery health, energy consumption, and vehicle status via IoT sensors.
3. Predict battery degradation and maintenance needs using ML models.

4. Optimize charging schedules against battery status, electricity demand, and station availability.
5. Recommend energy-efficient routes using GPS, traffic, and vehicle performance data.
6. Provide real-time dashboards for fleet managers.
7. Generate analytical reports on utilization, efficiency, and performance.
8. Improve productivity while reducing energy consumption and costs.

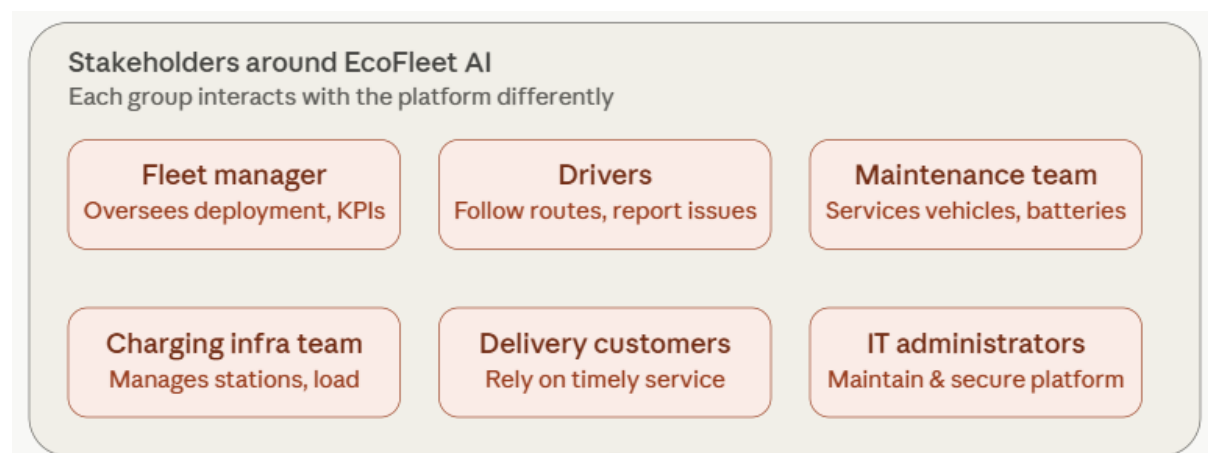
2.4 Scope

In scope: real-time IoT-based monitoring, predictive maintenance modelling, charging-schedule optimization, route optimization, manager dashboards and reporting, integration with existing telematics hardware.

Out of scope: physical charging-station construction, vehicle-to-grid (V2G) energy trading, autonomous driving, and national charging-policy design. EcoFleet AI is a software/analytics layer above existing EV and charging-network hardware, consistent with how current research frames the challenge IoT sensors aggregate real-time data from transport networks specifically to improve charging-demand forecasting and infrastructure decision-making, which is the layer this platform is built to occupy.

3. Stakeholders and Challenges

Below is the stakeholder ecosystem this platform must serve:



3.1 Roles and Responsibilities

Stakeholder	Role	Responsibilities
Fleet Manager	Operations lead	Assigns vehicles to routes/shifts, monitors KPIs, makes deployment decisions
Drivers	Vehicle operators	Operate vehicles, follow assigned routes, report anomalies
Maintenance Team	Technical service	Service vehicles, replace batteries/parts, respond to fault alerts
Charging Infra Team	Energy/infrastructure management	Manage stations, energy contracts, load balancing
Delivery Customers	End recipients	Depend on on-time pickup/delivery
IT Administrators	Platform upkeep	Maintain integrations, enforce data security policy
Sustainability Officers	Compliance	Track emissions reduction and regulatory adherence
Senior Management	Budget/ROI	Approve investment, evaluate returns

3.2 Major Challenges

1. **Information silos** : battery, GPS, and charging data typically sit in separate systems, so no one stakeholder sees the full picture.
2. **Reactive operations** :problems (low battery, breakdown) are usually discovered after they occur, not before.
3. **Charging congestion** : multiple vehicles competing for limited charging slots without intelligent scheduling causes queuing and downtime. This matches findings in EV-charging research, which frames the operational challenge around network-level optimization, parking-based strategies, demand-response mechanisms, and real-time scheduling systems needed to coordinate charging across a fleet.
4. **Trust and adoption** : drivers and dispatchers accustomed to manual workflows may be reluctant to follow AI-generated recommendations unless the system explains its reasoning transparently.

4. Analysis of the Current Approach

4.1 Existing System / Current Process

Most operators today rely on some combination of: basic GPS/telematics dashboards (location and speed only); manufacturer battery apps showing state-of-charge without fleet-wide forecasting; manual, spreadsheet-based charging schedules built on dispatcher experience; and traditional route-planning software optimized for distance/time, not energy.

4.2 Strengths

- Reliable, mature real-time location tracking.
- Familiar interfaces requiring little driver retraining.
- Low integration cost, since GPS hardware is already installed in most commercial EVs.

4.3 Limitations

- No predictive capability battery issues surface only as breakdowns.
- Routing is distance/time-optimized, not energy-optimized, so it frequently misses energy-cheaper paths and ignores charger placement.
- Charging scheduling is manual, with no awareness of real-time tariffs or station congestion.
- Data stays siloed across GPS, battery, and charging systems.
- Reporting is retrospective (weekly/monthly), delaying corrective action.

4.4 Gaps to Address

The core gap is the absence of an integrated, predictive intelligence layer. Existing routing-optimization research confirms this shortfall directly: many routing methods prioritize route length and duration but overlook EV-specific needs such as real-time charging and energy efficiency — exactly the gap EcoFleet AI closes by unifying battery-health prediction, charging optimization, and energy-aware routing into a single decision layer instead of three disconnected tools.

5. Proposed Software Solution

5.1 The Six Functional Modules

1. **IoT Data Ingestion** — collects battery telemetry, GPS, and charging-session data via MQTT/REST, buffered through a message queue (e.g., Kafka) for reliability during network fluctuation.
2. **Battery Health & Predictive Maintenance** — applies ML (regression, LSTM, or hybrid physics-informed models) to estimate State of Health and Remaining Useful Life. This mirrors current research showing that hybrid Scientific-Machine-Learning frameworks integrate domain knowledge with neural networks, producing more

interpretable and scientifically grounded battery-health predictions than pure black-box models and health-estimation surveys similarly stress that SoH/RUL prediction is essential because battery degradation behaves non-linearly.

3. **Charging Optimization** — recommends optimal charging windows and stations based on current state of charge, tariff schedules, and station occupancy, consistent with survey findings that effective charging management requires real-time scheduling and demand-response coordination across a network.
4. **Route Optimization** — computes energy-efficient routes using live traffic, elevation, and vehicle-specific energy models, treating charging stops as part of the route rather than an afterthought.
5. **Fleet Dashboard** — real-time web dashboard with role-based views for managers, drivers, and maintenance staff.
6. **Analytics & Reporting** — utilization, energy-efficiency, cost, and emissions-reduction metrics.

5.2 Functional Requirements

- Ingest and store real-time IoT sensor data.
- Predict battery SoH/RUL and generate maintenance alerts.
- Generate per-vehicle charging schedule recommendations.
- Compute and display optimized routes with charging stops.
- Provide role-based dashboard access.
- Generate periodic and on-demand reports.
- Send real-time alerts (low battery, maintenance due, route deviation).

5.3 Non-Functional Requirements

- **Performance:** dashboard updates within 2–3 seconds of new telemetry.
- **Scalability:** support growth from tens to thousands of vehicles without redesign.
- **Reliability:** 99.9% uptime target, graceful degradation on connectivity loss.
- **Security:** TLS encryption, role-based access control, secure API authentication.
- **Usability:** intuitive dashboard requiring minimal training.
- **Maintainability:** modular microservices, independently updatable.
- **Interoperability:** standard APIs for existing telematics/ERP integration.

5.4 Software Engineering Principles Applied

- **Modularity:** each module is an independent microservice with well-defined APIs.
- **Scalability:** containerized cloud-native deployment (Docker/Kubernetes) with a message-queue ingestion layer absorbing IoT traffic bursts.
- **Maintainability:** clear separation of concerns, versioned APIs, per-module test suites.
- **Reliability:** redundant pipelines and rule-based fallback logic if ML predictions are temporarily unavailable.
- **Usability:** role-specific dashboard views tested with real dispatchers and drivers.
- **Security:** encryption in transit/at rest, OAuth2 authentication, audit logging.

6. Supporting Mathematical Models

6.1 Battery Health & Predictive Maintenance

1. State of Health (SoH)

Formula

$$\text{SoH}(\%) = \frac{C_{\text{current}}}{C_{\text{rated}}} \times 100$$

Meaning

It measures how healthy the battery is compared to when it was new.

Where:

- **C_{current}** = Current battery capacity
- **C_{rated}** = Original battery capacity

Example

Original battery = **100 kWh**

Current capacity = **90 kWh**

SoH = $(90/100) \times 100 = \mathbf{90\%}$

This means the battery still has **90% of its original health**.

2. Remaining Useful Life (RUL)

Formula

$$RUL = \frac{SoH_{current} - SoH_{threshold}}{degradation\ rate}$$

Meaning

It predicts how many charging cycles the battery can still be used before replacement.

Example

Current SoH = 90%

Threshold = 80%

Battery degrades 0.05% per cycle

$RUL = (90 - 80) / 0.05$

$= 200$ cycles

Meaning the battery can be used for approximately **200 more charging cycles**.

3. Battery Stress Factor

Formula

$$Stress = w_1 \frac{\Delta T}{\Delta T_{max}} + w_2 \frac{DoD}{DoD_{max}} + w_3 \frac{CRate}{CRate_{max}}$$

Meaning

Measures how much stress the battery is experiencing.

Factors considered:

- Temperature
- Depth of Discharge (DoD)
- Charging speed (C-rate)

Higher stress means:

- Faster battery degradation
- Shorter battery life

4. AI Prediction Model

Formula

$$SoH(t + \Delta t) = f(...)$$

Meaning

Instead of using only formulas, an AI model (such as **LSTM**, Regression, or Neural-ODE) predicts future battery health using:

- Battery health history
- Temperature
- Charging cycles
- Charging rate
- Battery usage

This helps predict battery failure before it happens.

6.2 Charging Optimization

Goal:

Charge all vehicles at the lowest possible electricity cost.

1. Cost Minimization

Formula

Minimize

$$\sum Price(t) \times EnergyCharged(v, t)$$

Meaning

Choose charging times when electricity prices are lowest.

Example

Night:

₹6 per kWh

Day:

₹12 per kWh

AI schedules charging at night to reduce costs.

2. Charging Constraints

The AI ensures:

- Battery has enough charge before the next trip.
- Charging stations are not overloaded.
- Vehicles do not exceed their maximum charging speed.

3. Charging Time

Formula

$$t = \frac{(SoC_{target} - SoC_{current}) \times BatteryCapacity}{ChargingPower}$$

Meaning

Calculates how long charging will take.

Example

Current charge = 20%

Target = 80%

Battery = 60 kWh

Charger = 30 kW

Charging time

$$= (0.8 - 0.2) \times 60 / 30$$

$$= 1.2 \text{ hours}$$

4. Charging Station Ranking

Formula

$$Score = \alpha \left(\frac{1}{Distance} \right) + \beta \left(\frac{1}{WaitTime} \right) + \gamma \left(\frac{1}{Cost} \right)$$

Meaning

Ranks charging stations.

Higher score means:

- Closer station
- Less waiting
- Lower charging cost

AI selects the station with the highest score.

6.3 Route Optimization

Goal:

Find the route that uses the least battery energy.

Energy Consumption Formula

This formula estimates how much battery energy is used based on:

- Vehicle weight
- Road slope
- Air resistance
- Rolling resistance
- Vehicle speed
- Motor efficiency

Instead of choosing the shortest route, AI chooses the route that consumes the least energy.

Total Route Energy

$$TotalEnergy = \sum E_{segment}$$

The total battery energy is the sum of the energy required for every road segment.

AI chooses the route with the minimum total energy.

Safety Constraint

The battery should never drop below a safe level.

For example,

If safety margin = 15%

AI will insert a charging stop before the battery reaches 15%.

AI Algorithm

The system uses:

- **Energy-weighted Dijkstra Algorithm**
- **A* (A-Star) Search Algorithm**

These algorithms find the most energy-efficient route instead of simply the shortest one.

6.4 Fleet Analytics (KPIs)

1. Fleet Utilization

Formula

$$Utilization = \frac{OperatingHours}{AvailableHours} \times 100$$

Measures how efficiently vehicles are being used.

Example

Available = 10 hours

Used = 8 hours

Utilization = 80%

2. Energy Efficiency

Formula

$$\text{Energy Efficiency} = \frac{\text{Energy Consumed}}{\text{Distance}}$$

Measures battery consumption per kilometer.

Example

100 kWh used

400 km traveled

Efficiency = **0.25 kWh/km**

Lower values indicate better efficiency.

3. Cost Savings

Formula

$$\text{Savings} = (\text{OldRouteCost} - \text{NewRouteCost}) + (\text{OldChargingCost} - \text{NewChargingCost})$$

Calculates the money saved through optimized routing and charging.

4. CO₂ Emissions Avoided

Formula

$$\text{CO}_2 \text{ Avoided} = \text{EV Distance} \times \text{ICE Emission Factor}$$

Measures the reduction in carbon emissions by using EVs instead of petrol or diesel vehicles.

Example

Distance = 500 km

ICE emission factor = 0.18 kg CO₂/km

CO₂ avoided = $500 \times 0.18 = 90 \text{ kg CO}_2$

7. Justification of Methodology/Framework

Recommended: Agile (Scrum) with DevOps/MLOps integration

Here's the development cycle visualized:



7.1 Why Agile/Scrum over Waterfall or Spiral

- **Evolving requirements:** the battery-degradation and route-optimization models require iterative tuning as real data accumulates — a rigid Waterfall approach assumes fixed upfront requirements, which doesn't fit a system whose accuracy depends on continuous learning.
- **Stakeholder feedback loops:** fleet managers, drivers, and maintenance staff need to give continuous input on dashboard usability and recommendation accuracy; sprint reviews create a structured channel for this.
- **Incremental value delivery:** the six modules can ship as separate increments (dashboard and ingestion first, then predictive maintenance, then charging/route optimization), letting the operator benefit before the full system is complete.
- **DevOps/MLOps integration:** because the system depends on continuously retrained ML models and live IoT streams, a CI/CD pipeline for code plus automated retraining/validation for models keeps predictions accurate as fleet behavior and battery chemistries evolve.

7.2 How It Supports Development and Maintenance

- Two-to-three-week sprints let ML modules be refined against growing telemetry rather than frozen at initial design.
- CI/CD reduces the risk of regressions when updating any single microservice.
- Regular sprint demos to fleet managers build trust in AI recommendations, directly addressing the adoption-resistance challenge identified in Section 3.
- A product backlog lets new requirements (a new charging-station provider, a new telemetry format) be absorbed without a full re-planning cycle — important given how quickly the EV/charging research landscape itself is moving, reflected in the ongoing pace of published routing and charging-scheduling studies.

8. Expected Outcomes and Limitations

8.1 Expected Benefits

- Extended battery life and fewer unplanned failures through predictive maintenance.
- Lower energy and charging costs via tariff-aware scheduling.
- Shorter travel times and lower per-trip energy consumption through energy-aware routing.
- Higher vehicle availability and fleet utilization.
- Faster, better-informed decisions for fleet managers.
- Quantifiable emissions-reduction data for sustainability reporting.

8.2 How the Solution Addresses Identified Challenges

- Removes data silos by unifying battery, location, and charging data.
- Converts operations from reactive to proactive through predictive alerts.
- Reduces charging-station congestion through intelligent scheduling.
- Builds stakeholder trust through transparent, explainable dashboard recommendations.

8.3 Risks, Implementation Challenges, and Limitations

- **Data quality dependency:** prediction accuracy depends on sensor completeness; noisy or missing IoT data degrades model performance.
- **Integration complexity:** inconsistent data formats across manufacturers' telematics hardware complicate ingestion.
- **Model drift:** battery and routing models need continuous retraining as usage patterns and infrastructure evolve — without sustained MLOps investment, accuracy decays.
- **Adoption resistance:** drivers/dispatchers used to manual processes may distrust automated recommendations initially.
- **Infrastructure dependency:** optimization quality is bounded by actually available charging infrastructure in a region.
- **Upfront cost:** IoT retrofitting for older vehicles and cloud setup require initial investment.

8.4 Future Enhancements

- Vehicle-to-grid (V2G) integration to sell surplus energy back to the grid during idle periods.
- Weather-aware energy/route models (temperature strongly affects EV efficiency).
- Extending predictive maintenance to motor and tire wear, not just batteries.
- Driver-behavior coaching to further cut energy consumption.
- Renewable-energy-aware charging that prioritizes charging when solar/wind supply is high.

9. Conclusion

EcoFleet AI addresses a structural gap in current EV fleet management: the absence of a single platform that connects battery health, charging strategy, and route planning through predictive intelligence rather than treating them as separate concerns. By combining IoT-based real-time monitoring, machine-learning-driven predictive maintenance, optimization-based charging and routing, and a modular, Agile-developed architecture, the platform is positioned to reduce operating costs, extend battery life, improve fleet utilization, and support measurable emissions reduction. Its success, however, is conditional — it depends on clean sensor data, ongoing model retraining, and genuine stakeholder trust in AI-driven recommendations. These are manageable risks with the right MLOps discipline and change-management approach, making EcoFleet AI a technically sound and commercially justifiable investment for logistics, transit, and delivery fleets transitioning to electric vehicles.

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